KAIZEN MATHS

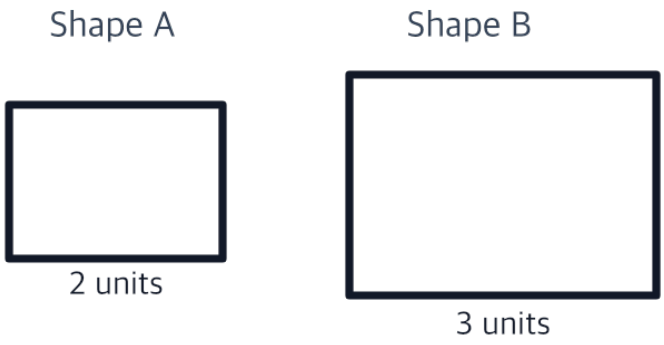
Lesson starter

Scale Drawings and Similar Shapes

Area & Volume Ratios

Name _____

SA1



Two similar shapes have length ratio 2:3. The area of the smaller shape is 20 cm². Find the area of the larger shape.

GEO-9EE4-L4 · SA

Show your working

KAIZEN MATHS

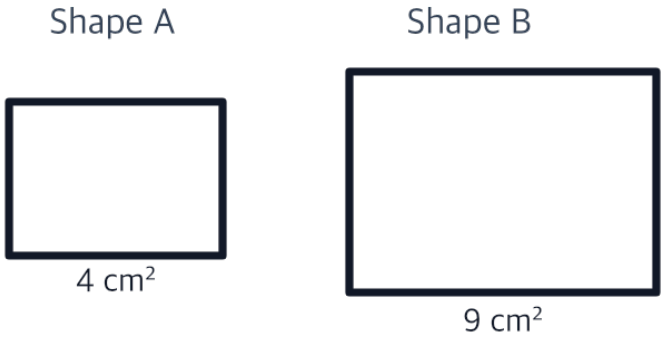
Lesson starter

Scale Drawings and Similar Shapes

Area & Volume Ratios

Name _____

SB1



Two similar shapes have area ratio 4:9. Find their length ratio.

GEO-9EE4-L4 · SB

Show your working

KAIZEN MATHS

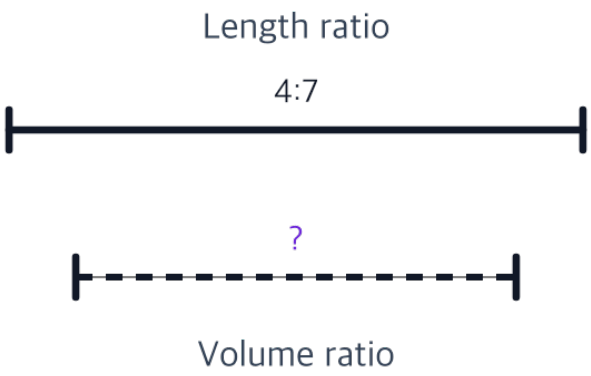
Lesson starter

Scale Drawings and Similar Shapes

Area & Volume Ratios

Name _____

SC1



Two similar solids have length ratio 4:7. The smaller solid has volume 128 cm^3. Find the volume of the larger solid.

GEO-9EE4-L4 · SC

Show your working

KAIZEN MATHS

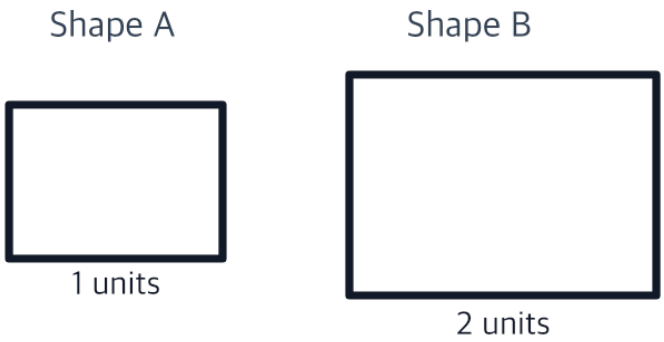
Lesson starter

Scale Drawings and Similar Shapes

Area & Volume Ratios

Name _____

SD1



Two similar shapes have length ratio 1:2. The area of the smaller shape is 3 cm^2. Find the area of the larger shape.

GEO-9EE4-L4 · SD

Show your working

KAIZEN MATHS

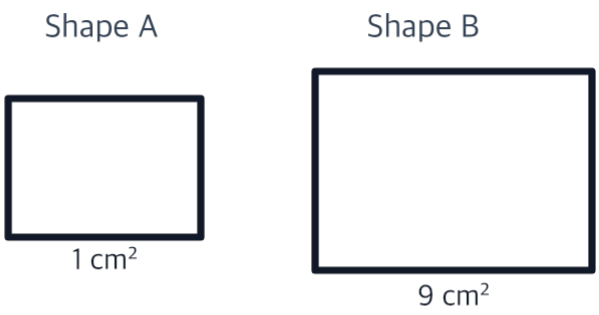
Exit ticket

Scale Drawings and Similar Shapes

Area & Volume Ratios

Name

EA1



Two similar shapes have area ratio 1:9. Find their length ratio.

Show your working

KAIZEN MATHS

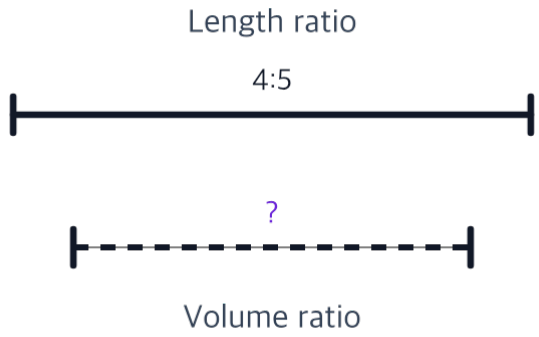
Exit ticket

Scale Drawings and Similar Shapes

Area & Volume Ratios

Name

EB1



Two similar solids have length ratio 4:5. The smaller solid has volume 192 cm^3. Find the volume of the larger solid.

Show your working

KAIZEN MATHS

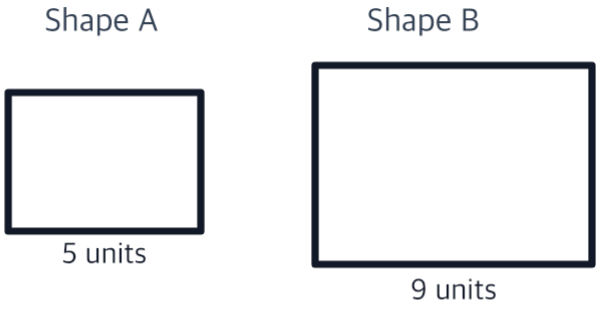
Exit ticket

Scale Drawings and Similar Shapes

Area & Volume Ratios

Name

EC1



Two similar shapes have length ratio 5:9. The area of the smaller shape is 150 cm^2. Find the area of the larger shape.

Show your working

KAIZEN MATHS

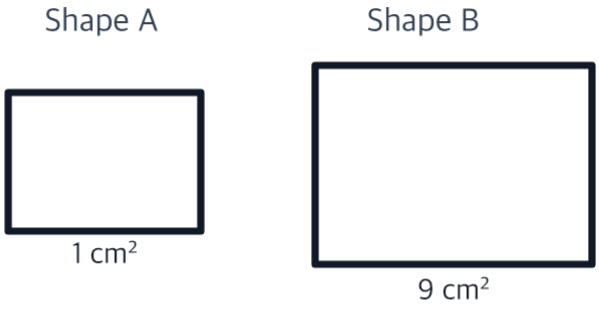
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Scale Drawings and Similar Shapes

Area & Volume Ratios

Name

EA1



Two similar shapes have area ratio 1:9. Find their length ratio.

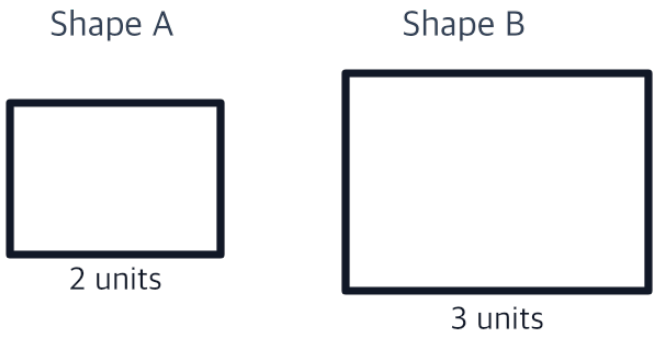
Show your working

Teacher answers & working

Scale Drawings and Similar Shapes · Area & Volume Ratios

GEO-9EE4-L4 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1



Two similar shapes have length ratio 2:3. The area of the smaller shape is 20 cm². Find the area of the larger shape.

Answer:

45 cm²

Working:

Area ratio is the square of the length ratio.

Length ratio = 2:3.

Area ratio = $2^2:3^2 = 4:9$.

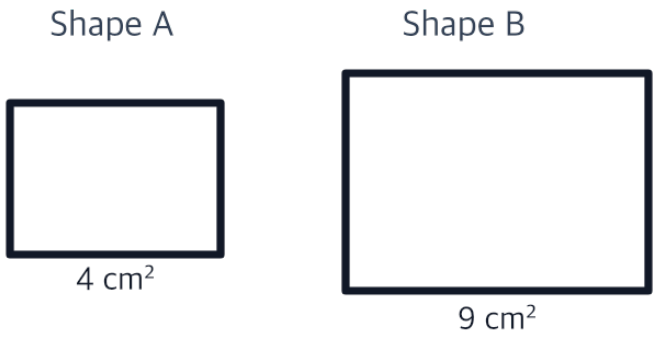
Larger area = $20 \times 9 \div 4 = 45$ cm².

Teacher answers & working

Scale Drawings and Similar Shapes · Area & Volume Ratios

GEO-9EE4-L4 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1



Two similar shapes have area ratio 4:9. Find their length ratio.

Answer:

2:3

Working:

Area ratio is the square of the length ratio.

Take the square root of each part of the area ratio.

$\sqrt{4}:\sqrt{9} = 2:3$.

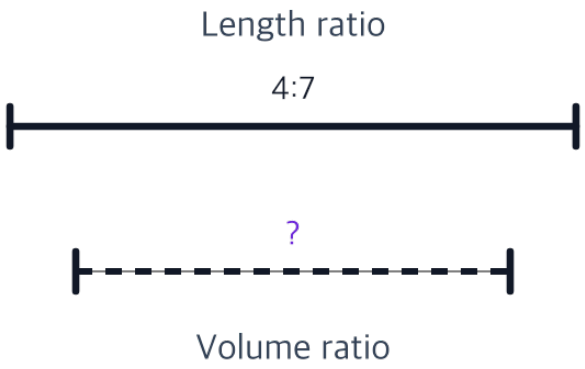
So the length ratio is 2:3.

Teacher answers & working

Scale Drawings and Similar Shapes · Area & Volume Ratios

GEO-9EE4-L4 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SC1



Two similar solids have length ratio 4:7. The smaller solid has volume 128 cm³. Find the volume of the larger solid.

Answer:

686 cm³

Working:

Volume ratio is the cube of the length ratio.

Length ratio = 4:7.

Volume ratio = $4^3:7^3 = 64:343$.

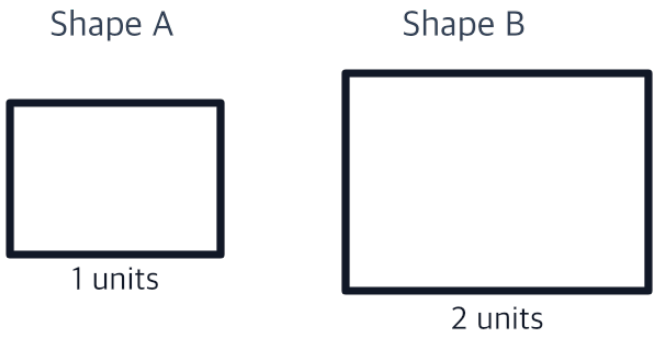
Larger volume = $128 \times 343 \div 64 = 686$ cm³.

Teacher answers & working

Scale Drawings and Similar Shapes · Area & Volume Ratios

GEO-9EE4-L4 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SD1



Two similar shapes have length ratio 1:2. The area of the smaller shape is 3 cm². Find the area of the larger shape.

Answer:

12 cm²

Working:

Area ratio is the square of the length ratio.

Length ratio = 1:2.

Area ratio = $1^2:2^2$ = 1:4.

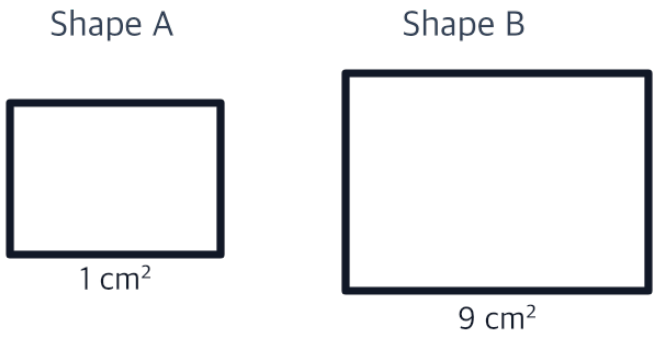
Larger area = 3 × 4 ÷ 1 = 12 cm².

Teacher answers & working

Scale Drawings and Similar Shapes · Area & Volume Ratios

GEO-9EE4-L4 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1



Two similar shapes have area ratio 1:9. Find their length ratio.

Answer:

1:3

Working:

Area ratio is the square of the length ratio.

Take the square root of each part of the area ratio.

$\sqrt{1}:\sqrt{9} = 1:3$.

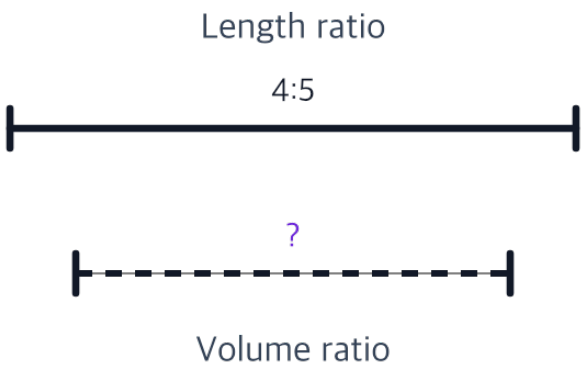
So the length ratio is 1:3.

Teacher answers & working

Scale Drawings and Similar Shapes · Area & Volume Ratios

GEO-9EE4-L4 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1



Two similar solids have length ratio 4:5. The smaller solid has volume 192 cm^3. Find the volume of the larger solid.

Answer:

375 cm^3

Working:

Volume ratio is the cube of the length ratio.

Length ratio = 4:5.

Volume ratio = $4^3:5^3$ = 64:125.

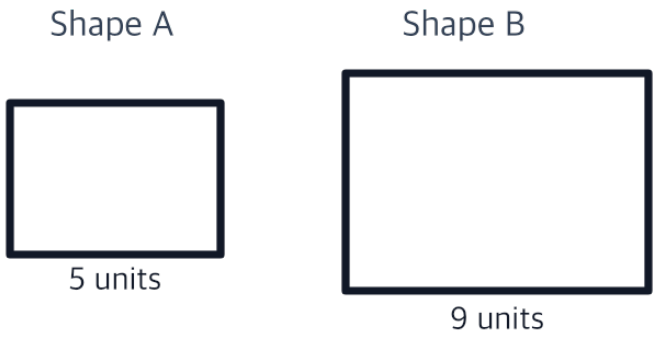
Larger volume = 192 × 125 ÷ 64 = 375 cm^3.

Teacher answers & working

Scale Drawings and Similar Shapes · Area & Volume Ratios

GEO-9EE4-L4 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EC1



Two similar shapes have length ratio 5:9. The area of the smaller shape is 150 cm². Find the area of the larger shape.

Answer:

486 cm²

Working:

Area ratio is the square of the length ratio.

Length ratio = 5:9.

Area ratio = 5^2:9^2 = 25:81.

Larger area = 150 × 81 ÷ 25 = 486 cm².